

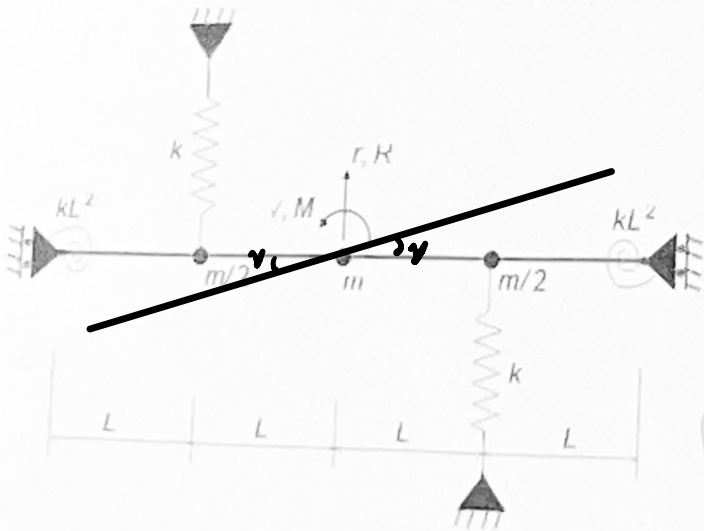
Problem to solve:

1. Consider: $\rho A \frac{\partial^2 v}{\partial t^2} + EI \frac{\partial^4 v}{\partial z^4} = 0$ Show that, for $v(z, t) = V(z) \cos(\omega t - \alpha)$ with

$$V = C_1 \sinh(\lambda z) + C_2 \cosh(\lambda z) + C_3 \sin(\lambda z) + C_4 \cos(\lambda z)$$

Show that it must be that: $\lambda^4 = \frac{\rho A \omega^2}{EI}$

2. Using Lagrange's equations, find the equation of motion for this system:



$$1. \frac{\partial^2 v}{\partial t^2} = -\omega^2 V \cos(\omega t - \alpha) = -\omega^2 v$$

$$\frac{\partial^4 v}{\partial z^4} = \frac{\partial^4 V}{\partial z^4} \cos(\omega t - \alpha)$$

$$= \lambda^4 C_1 \sinh(\lambda z) + \lambda^4 C_2 \cosh(\lambda z) + \lambda^4 C_3 \sin(\lambda z) + \lambda^4 C_4 \cos(\lambda z)$$

$$= \lambda^4 V \cos(\omega t - \alpha) = \lambda^4 v$$

$$\therefore -\rho A \omega^2 v + EI \lambda^4 v = 0$$

$$\therefore \lambda^4 = \frac{\rho A \omega^2}{EI}$$

$$2. \quad T = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} \cdot \frac{1}{2} m \cdot (\dot{r} - L\dot{v})^2 + \frac{1}{2} \cdot \frac{1}{2} m \cdot (\dot{r} + L\dot{v})^2$$

$$= m\dot{r}^2 + \frac{1}{2} m L^2 \dot{v}^2$$

$$U = \frac{1}{2} k \cdot (r - Lv)^2 + \frac{1}{2} k \cdot (r + Lv)^2 + \frac{1}{2} k L^2 (v)^2 + \frac{1}{2} k L (v)^2$$

$$= kr^2 + kL^2 v^2 + kL^2 v^2 = kr^2 + 2kL^2 v^2$$

$$\therefore L = T - U$$

$$\frac{\partial L}{\partial r} - \frac{d}{dt} \frac{\partial L}{\partial \dot{r}} = 0$$

$$\therefore \frac{d}{dt} \frac{\partial T}{\partial \dot{r}} - \frac{\partial T}{\partial r} + \frac{\partial U}{\partial r} + \frac{\partial Q}{\partial r} = R$$

for r, R

$$2m\ddot{r} + 2kr = R$$

for v, M

$$mL^2\ddot{v} + 3kL^2 v = M$$

$$\therefore m \begin{bmatrix} 2 & 0 \\ 0 & L^2 \end{bmatrix} \begin{bmatrix} \ddot{r} \\ \ddot{v} \end{bmatrix} + k \begin{bmatrix} 2 & 0 \\ 0 & 4L^2 \end{bmatrix} \begin{bmatrix} r \\ v \end{bmatrix} = \begin{bmatrix} R \\ M \end{bmatrix}$$